Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/15/2024
PI Name:	Davis, Catherine M. Ph.D.		
Project Title:	VNSCOR: Mechanisms of Radiation-Induced Changes in Sustained Attention and Social Processing (80NSSC22K0022)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NON-PROFIT	**Phone:**	301-400-4596
Organization Name:	Henry M. Jackson Foundation		
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Zip Code:	20817-1891	**Congressional District:**	8
Comments:	Campus address (Jan 2022): Department of Pharmacology and Molecular Therapeutics, Uniformed Services University of the Health Sciences, 4301 Jones Bridge Road, Bethesda, MD 20814. NOTE: PI formerly at Johns Hopkins University; moved to Henry M. Jackson Foundation for the Advancement of Military Medicine in fall 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	12/14/2021	**End Date:**	03/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	End date changed to 03/31/2026 per the PI (Ed., 4/21/26). End date changed to 12/13/2025 per L. Juliette (Ed., 12/19/24). End date corrected to 12/13/2024 per V. Lehman/JSC Grants Office (Ed., 2/28/2024). End date entered as 12/13/2025 per NSSC information (Ed., 10/26/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert Ph.D. (Johns Hopkins University) Robinson, Siobhan Ph.D. (Hamilton College)		

Grant/Contract No.:	80NSSC22K0022
Performance Goal No.:	
Performance Goal Text:	
Task Description:	[Ed. note Jan 2022: Continuation with same Principal Investigator (PI) Dr. Catherine Davis, of "VNSCOR: Mechanisms of Radiation-Induced Changes in Sustained Attention and Social Processing," grant #80NSSC18K1080 when PI was at Johns Hopkins University. See also project, "VNSCOR: Responses of the Nervous System to Chronic, Low Dose Charged Particle Irradiation" (Principal Investigator (PI): Greg Nelson)] NELSON/DAVIS VIRTUAL NASA Specialized Center of Research (NSCOR): The project is organized as 5 large experimental campaigns to quantify responses for an interrelated set of central nervous system (CNS) outcome measures in mice to acute and protracted exposures to protons at a dose of 0.5 Gy and sham controls; acute and protracted exposures to 0.25 and 0.5 Gy of charged particles; and acute and protracted exposures to 0.5 and 1.5 Gy of gamma rays. This research builds on previous studies that demonstrated that proton and HZE (high charge energy) exposures result in individual differences in deficits in sustained attention, but more general deficits in recognition memory. This current project is combined with "Responses to the Nervous System to Chronic, Low Dosed Charged Particle Irradiation" (PI: Nelson) in order to explore if these effects are LET (linear energy transfer)-dependent for 16O ions, add a relatively understudied, but important, ion (4He), and examine CNS effects in whole animals following fractionated exposures, and the interaction of other space flight factors (e.g., sleep fragmentation). Revised Specific Aims: Aim 1a: Effects of protracted exposure to five-ion GCR [galactic cosmic rays] sim (1H, 4He, 28Si, 16O, 56Fe) Aim 1b: Effects of protracted exposure to protons only Aim 1c: Acute exposure to 4He (250 MeV/n) Aim 2a: Neuronal activation and molecular markers following radiation Aim 2b: Chemogenetic silencing of mPFC subregions Specific Aims: 1) Determine the effects of acute, single 16O and 4He ion exposures on sustained attention, social odor recognition memory, and social dominance. (This aim has been modified in order to integrate with Nelson project); 2) Determine the effects of a fractionated exposure on sustained attention and recognition memory in comparison to the effects of single ion exposures on these measures; 3) Determine the effects of circadian disruptions and sleep fragmentation on sustained attention and recognition memory following radiation exposure (This aim has been modified in order to integrate with Nelson project); 4) Examine the underlying mechanisms of these deficits using immunohistochemical and pharmacogenetic procedures. ORIGINAL PROPOSAL DESCRIPTION: Assessing the biological consequences of living in the space radiation environment represents one of the highest priority areas of NASA research. Of critical importance is the need for an assessment of the vulnerabilities of the central nervous system (CNS) leading to functional neurobehavioral changes during long-term space missions, and the development of effective countermeasures to such risks. The present proposal addresses this need via the application of an animal model to 1) determine the effects of acute, single 16O and 4He ion exposures on sustained attention, social odor recognition memory, and social dominance; 2) determine the effects of a fractionated exposure on sustained attention and recognition memory in comparison to the effects of single ion exposures on these measures; 3) determine the effects of circadian disruptions and sleep fragmentation on sustained attention and recognition memory following radiation exposure; and 4) examine the underlying mechanisms of these deficits using immunohistochemical and pharmacogenetic procedures. Prior research has 1) identified rats that are sensitive to radiation-induced deficits in sustained attention and 2) shown that acute, single ion exposures alter social motivation and social odor recognition memory. The current proposal will determine how the immediate effects of irradiation impact subsequent neurobehavioral deficits by assessing various behavioral, physiological, and neurobiological markers of radiation exposure at early and later time points post-exposure. Groups of animals will be trained on a rodent version of the human psychomotor vigilance test, exposed to radiation, and then tested for both social processing and sustained attention deficits following exposure; subsets of rats will be implanted with biotelemetry devices to measure fluctuations in spontaneous locomotor activity and body temperatures following radiation exposure and changes in sleep or circadian disruptions. Individual variations in these behavioral, physiological, or neurobiological responses following radiation will be assessed. Likely mechanisms of damage to the CNS following radiation exposure will be examined using brain tissue, in addition to a pharmacogenetic technique to determine the severity of radiation-induced neurobehavioral deficits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the current project will be used to understand how radiation exposure affects the central nervous system to induce deficits in neurobehavioral function.
	In general, our current results suggest several important things, including: 1) behavioral performance, especially recognition memory, is variable following radiation exposure and that longitudinal testing is required to see this variability and understand the implications it has for operational performance; 2) that female rats are not necessarily more resilient to the central nervous system (CNS) effects of space radiation exposure, as suggested by previously published work with mice, but that the timeline of deficits following exposure could be sex-specific; 3) not all cognitive domains are affected in the same manner following exposure. This fact has been shown in work from other labs studying the CNS effects of space radiation exposure, and our current data lend greater support to this idea. In terms of astronaut performance, the results from the studies completed thus far suggest that acute and chronic, single ion exposure, acute multi-ion exposure, and chronic multi-ion exposure result in deficits in social recognition memory

Task Progress:	and sustained attention, potential increases in anxiety-like behavior in female rats, but limited changes in social interaction. The multi-ion exposures induce deficits in social memory that appear as early as 7 days following radiation exposure, and these early deficits appear to be independent of dose rate and sex, such that the deficits are equivalent between the acute and chronic exposure groups. These deficits vary over the post-radiation testing period. All exposures induce social recognition deficits early in the post-exposure period, which suggests that measures to limit or lower the exposure dose could be beneficial for astronauts on long duration missions. For example, we previously reported no early (7 day) social recognition memory deficits following 5 cGy 4He exposure, and this same dose also induced limited changes in rodent psychomotor vigilance test (rPVT) performance parameters. Further, these early deficits appear to be more variable for lower exposure doses, findings that suggest that lower exposure doses do not necessarily induce permanent changes in this cognitive domain. In contrast, we've reported that higher doses (i.e., 10 cGy and above) appear to induce variable deficits in social recognition that show no recovery at the time points we tested. If countermeasures and/or shielding can reduce the dose of galactic cosmic rays (GCR) to which an astronaut is exposed, radiation-induced deficits in recognition memory could possibly be kept to a minimum during the mission. However, it is likely that exposure to these low doses would still result in long-term recognition memory deficits that develop after the mission. For sustained attention, performance on the rPVT was sensitive to radiation exposure in both sexes. As we've reported previously, proton exposure results in limited changes in overall rPVT performances based on mean daily performances. We have extended these findings to both acute and chronic proton exposures, in addition to female rats. In our previous work, protons did induce individual changes in performance and we identified "sensitive" subjects. These analyses are ongoing to determine if these individual differences are found for proton exposure in male and female rats. GCR-induced changes appear at the group level. When radiation exposure does induce changes in sustained attention, these deficits develop and worsen over time, with no evidence of recovery. While this provides time for countermeasure administration, this continued decrease in sustained attention performance is concerning for long-duration spaceflight. These data also suggest that sustained attention performance would be more impaired near the end of a mission, which is problematic for important end-of-mission tasks, especially if quick, accurate responding is needed.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Davis CM, Elliott VE, Smith J. "Sleep disruption impairs sustained attention in food-restricted rats using a food-reinforced rodent psychomotor vigilance test." Nat Sci Sleep. 2024 Nov 9:16:1771-7. Epub 2024 Nov 9. https://doi.org/10.2147/NSS.S475672 ; PMID: 39539357; PMCID: PMC11559244 , Nov-2024